very destructive. When earth material moves down a hillside as a fluid-like mass, it is called an **earth flow**. These flows typically occur in humid areas on steep slopes with thick, clay-rich soil that becomes saturated with water during storms.



Figure 4.7 Debris flow

A mudflow

Mudflow is a liquid mass of soil, rock debris and water that moves quickly down a well defined channel. They occur most often in mountainous semiarid environments. A mudflow originating on a volcanic slope is called a **lahar**.

Debris avalanche

The deadliest type of debris flow is **the debris avalanche**. It is a rapidly churning mass of rock debris, soil, water, and air that moves down steep slopes. The trapped air may increase the speed of an avalanche by acting as a cushion between the debris and the underlying surface

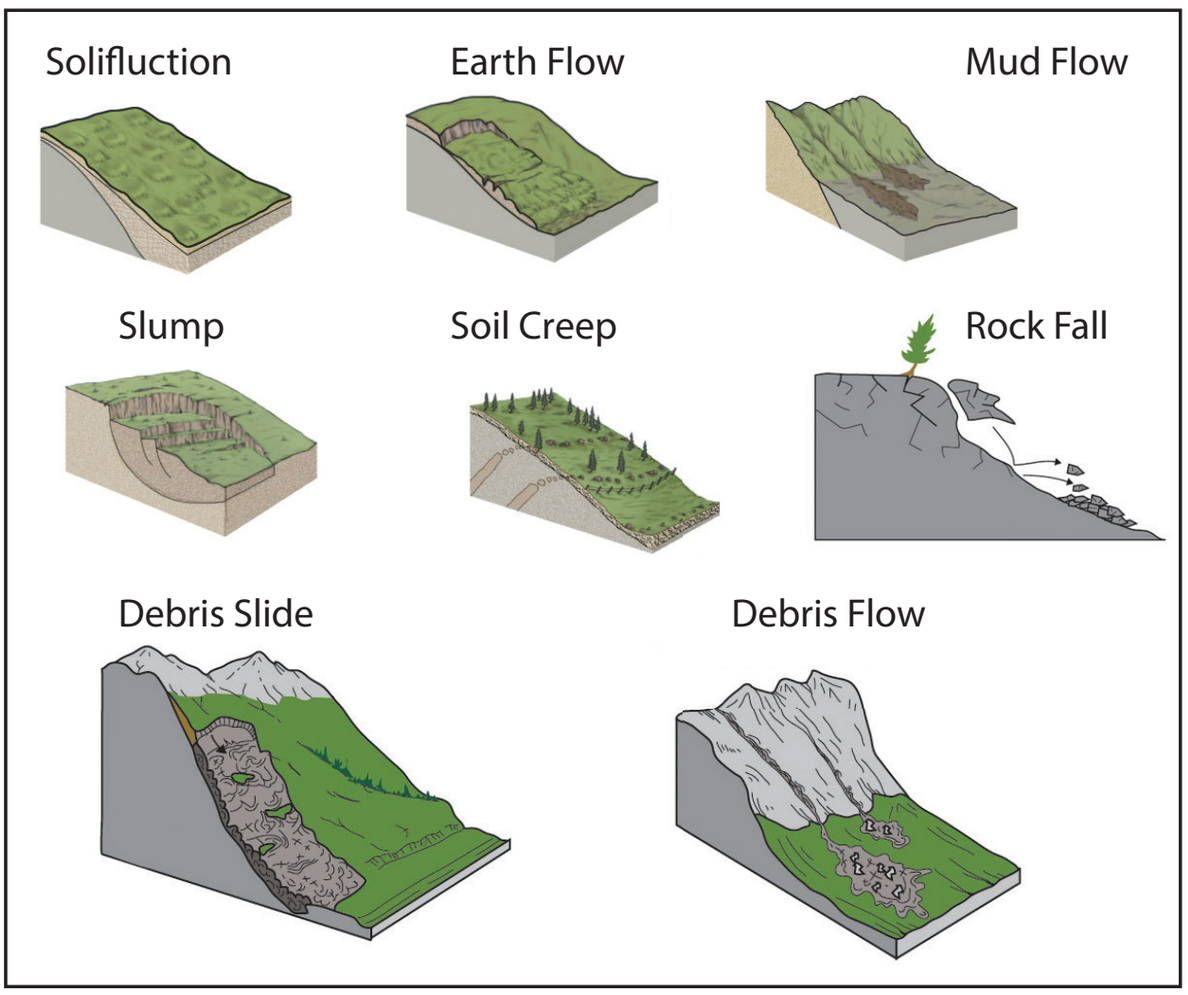


Figure 4.8 Mass Movement

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